

M-Lean: An end-to-end development framework for predictive models in B2B scenarios

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ABSTRACT

Context: The need for business intelligence has led to advances in machine learning in the business domain, especially with the rise of big data analytics. However, the resulting predictive systems often fail to maintain a satisfactory level of performance in production. Besides, for predictive systems used in business-to-business scenarios, user trust is subject to the model performance. Therefore, the processes of creating, evaluating, and deploying machine learning systems in the business domain need innovative solutions to solve the critical challenges of assuring the quality of the resulting systems.

Objective: Applying machine learning in business-to-business situations imposes specific requirements. This paper aims at providing an integrated solution to businesses to help them transform their data into actions.

Method: The paper presents *MLean*, an end-to-end framework, that aims at guiding businesses in designing, developing, evaluating, and deploying business-to-business predictive systems. The framework employs the Lean Startup methodology and aims at maximizing the business value while eliminating wasteful development practices.

Results: To evaluate the proposed framework, with the help of our industrial partner, we applied the framework to a case study to build a predictive product. The case study resulted in a predictive system to predict the risks of software license cancellations. The system was iteratively developed and evaluated while adopting the management and end-user perspectives.

Conclusion: It is concluded that, in industry, it is important to be aware of the businesses requirements before considering the application of machine learning. The framework accommodates business perspective from the beginning to produce a holistic product. From the results of the case study, we think that this framework can help businesses define the right opportunities for applying machine learning, developing solutions, evaluating the effectiveness of these solutions, and maintaining their performance in production.

1. Introduction

Big data is increasingly becoming a major organizational asset for all sizes of industries. The volume of the data generated in industry, from cloud-based systems, management solutions, and so forth, is growing enormously [1]. This exponential growth of data presents new challenges of how to make sense of the data, discover hidden trends in it, and employ this information to improve business operations. The primary objective of using big data in industry is to maintain cost-effective business processes. By the appropriate interpretation of such big data, businesses can create more efficient risk management systems and derive value in areas such as decision making, product development, and

improvement of customer experience. As a result, research focusing on big data solutions is becoming an essential requirement for future industrial applications.

Consequently, for the last few years, there has been a considerable amount of research focusing on big data analytics in industry. Some research focuses on defining the current challenges of big data [1–5]. Other research presents novel solutions that utilize big data in solving business challenges [6–8]. Most of these solutions exploit machine learning (ML) techniques to resolve business problems. The massive rise of ML techniques opened a wide range of possibilities in developing predictive models that help in many categories of business problems such as marketing [9,10], sales [11], customer churn [6], etc. However, most

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of the solutions presented in the literature are tailored to solve domain-specific problems. The topic of providing a systematic framework for utilizing big industrial data receives minimal attention in the literature. Based on our survey, none of the existing studies has focused on guiding businesses to define possible opportunities for exploiting their data to build predictive models.

Therefore, in this paper, we propose *MLean* which is a framework to steer businesses to transform their data into actions through building Business-to-Business (B2B) [12] predictive systems. The framework employs the Lean Startup methodology [13] to maximize the business value of the developed systems while eliminating wasteful development practices. To eliminate uncertainty coupled with the application of ML in industry, *MLean* applies different types of research designs through a sequence of phases. Precisely, since the question of the usefulness of the final system constitutes a significant source of uncertainty, the framework applies an exploratory research phase in the beginning to qualify the business value of the final model based on insights collected from real users and business leaders. Afterward, the framework conducts improving research in subsequent phases to test and maintain the business vision about the final model. Moreover, to sustain an adequate level of model performance, the framework applies various methods for data collections to obtain feedback from different stakeholder groups throughout the development and deployment phases. The primary contributions of this research can be summarized as follows:

- An end-to-end development framework is proposed to develop, evaluate, and deploy predictive products in business domains. It is argued that this is the first such end-to-end lifecycle process for data-intensive application development for B2B scenarios where a rich cross-section of stakeholders is actively engaged in the process. It is also argued that such engagement is essential if we hope to realize successful product lines.
- With the help of our industrial partner, IBM, we have applied our framework to a case study to build a B2B predictive product that predicts software license cancellations. That is, we undertake and report on an initial evaluation of the feasibility of the approach.

This paper is structured as follows: Section 2 presents the related work. The study scope and the research questions are represented in Section 3. While Section 4 discusses the research methodology, Section 5 introduces the overall architecture of the *MLean* framework. The application of the *MLean* to a case study in the IT industry is presented in Section 6. Section 7 discusses the lessons learned from the case study, reflects the cost of the implementation, and presents the threats to validity of the case study, while Section 8 concludes the paper.

2. Related work

Giving the essential role of ML systems in the business domain, there is a need to address the challenges that ML components bring into software systems. The primary focus of this research is to present a systematic structure for developing, evaluating, and deploying predictive systems in B2B scenarios. Therefore, we survey previous work that looks at the intersection between ML and software engineering. We classify the related work into three categories. The first category considers the application of ML in software engineering [14–17]. The applications include utilizing ML techniques in predicting software fault and defects [14,16] recommending process model [17], and estimating development effort [15]. Although this category aims at using ML techniques to optimize the process of creating software systems, none of these efforts looks at the challenges of using ML components as a part of the software systems.

As for the second category, since we present predictive models as a new class of requirements engineering problems, we survey existing work that combines the domains of ML and requirements engineering. We found that, over the last decade, many researchers have used ML models in analyzing the requirements for different software systems

[18–20]. Research [18] employs supervised learning approaches to classify requirements as functional and non-functional requirements. Authors in [19] use ML to prioritize software requirements by combining the stakeholders preferences with the requirements ordering. Also, authors in [20] present an automated ranking system for managing potential risks. However, in contrast to our work, none of the existing studies have focused on eliciting the requirements of ML applications themselves.

The third category aims at addressing the challenges in developing ML systems [21–24]. Previous work [21,22] states that distributed systems are required for an end-to-end ML pipeline. Authors in [22] propose an open source distributed ML library for scalable implementation of standard ML techniques. Researchers in [23] presented a system to optimize end-to-end ML systems, while authors in [24] produced a system to manage ML models. Although these studies focus on offering solutions to ease the process of creating ML models, none of these solutions tried to consider the perspective of business management and the end users. These efforts did not address the challenges of identifying opportunities to improve business processes using ML. Moreover, most of these solutions did not consider the requirements enforced by the business domain.

In contrast to the previous work, we offer an integrated ML framework that cooperates a broader range of insights, as we think that developing, evaluating, and maintaining ML products should not only consider the perspective of data science team. Instead, the real worlds input must be accommodated as well. Perhaps, the closest works we can find are life-cycle descriptions for ML development which describe the process from the data scientists viewpoint.¹ However, these life-cycles are focused on technical aspects and ignore the essential roles of business leaders, the marketplace, end-users and other stakeholders required to produce a holistic product line rather than just a ML algorithm. Hence, it is believed that this work is unique in its scope in providing an end-to-end life-cycle process for data-intensive, commonly ML-based, B2B applications.

3. Study scope

In the last few years, a considerable amount of research has taken place to apply ML techniques to industrial problems [6–12]. However, most of this work focuses on solving domain-specific problems. Therefore, it is hard to generalize these solutions to a broader range of applications. In this research, we start by asking some questions that formed the basis of our study:

RQ1: *What extent of research has been done to create an end-to-end framework for building predictive models in B2B scenarios?*

RQ2: *What are the design decisions required to create an end-to-end framework for building predictive models in B2B scenarios?*

RQ3: *What actions must be taken to apply and evaluate the effectiveness of such frameworks in industry?*

To answer the research questions, we limit our literature survey to the research that has been undertaken during the last five years. Regarding the firmographics variables [25], we focus on international organizations that have more than 10,000 employees. Each organization has its data science team, which is responsible for managing and analyzing the business datasets. Consequently, technical experience in predictive analytics and ML is expected within this team, since they work to apply ML with the aim of providing business solutions.

Additionally, this research primarily focuses on predictive systems in B2B [12] situations where both the seller and the buyer are organizations. Data in B2B scenarios is usually more complex than the data in Business-to-Consumer situations. In B2B environments, companies build

¹ <https://docs.microsoft.com/en-us/azure/machine-learning/team-data-science-process/lifecycle>.

long-term relationships with their customers, which results in data collected from diverse sources such as historical records and social media analytics. Therefore, models built to process such data must be able to handle this level of data complexity. Also, predictions generated for B2B scenarios are often followed by business decisions. Therefore, a level of interpretability [12] is required to gain more confidence about the following business decision.

4. Research methodology

In this section, we discuss the undertaken steps of our research methodology. Based on our literature survey and interaction with industry, we perceive that there are many challenges in applying ML in the business domain [5,9,26]. As a result, we employed an action research approach [27] intending to resolve some of these challenges. This research involves a longitudinal case study applied, with the help of our industrial partner, over more than nine months. A group of two researchers (the first two authors) worked to coordinate the application of the proposed framework. In our approach settings, the research team was treated as a part of one stakeholder group (i.e., the data science team) and not seen as researchers by the other stakeholder groups. This allowed the research team to iteratively integrate their theories within practice and continuously validate their hypotheses based on the experience gained throughout the case study.

The rationale behind using the action research approach originated from the correlation between the general elements of the action research approach and the primary principles of the proposed framework in two main points. First, action research shapes a collaborative process between researchers and different stakeholders in a given context. Second, it enforces a process of critical inquiry and reflective learning as a part of the research.

Therefore, as a start, the two researchers surveyed previous efforts in the literature to establish a framework of reference for the research and to answer RQ1 (Section 3). Then, we developed the research hypotheses for our study that aims at answering the research questions stated in Section 3.

As mentioned in Section 2, communication with different stakeholder groups is an essential factor that influences successful development of any ML model. Hence, we formulate our first hypothesis that seeks to answer RQ2 as follows:

H1: Effective interaction with business leaders and end users is positively connected with successful product lines. That is the more interaction, the more successful the predictive model.

Also, since a significant gap is noticed in research that focuses on analyzing the requirements of predictive models in industry, we formulate our second hypothesis that seeks to answer RQ3 as follows:

H2: Addressing the business requirements for the predictive model is positively connected with a satisfactory level of performance in production.

Subsequently, we proceed with the research by designing the framework (Section 5). The framework creates a continuous interaction with the stakeholders in which the business domain is recurrently investigated for possible opportunities for applying ML solutions. To facilitate the application of the proposed framework, the framework was designed as a sequence of phases. Along with the design of each phase, the framework outlines the application of each phase by defining the phase objective, the research questions, and the recommended methods for data collection. To apply the framework, the framework users can apply the suggested practices in each phase to collect data, answer the research questions, and progress to the subsequent phase. This should ease the integration of the framework in the business workflow. The final output of the framework is a B2B predictive model which has been trained using business data and has proven its effectiveness in building user trust. To validate our hypotheses, we apply the framework to a real case study (Section 6), in which the two researchers followed the framework design to build a system to predict software license cancellations.

Regarding data collection, several data collection methods were employed through the application of the MLean framework, including interviews (Section 6.2), participant observation (Section 6.3), group meetings (Section 6.4), and analysis of historical data records (Section 6.5). The diversity of the data method collection helped with data triangulation [28] and eliminated the risk of systematic biases. The rationale for choosing each data collection method is further elaborated in the following section along with the description of each phase of the framework.

5. Proposed framework design

This section describes the overall structure of the MLean framework. A high-level component overview of the framework is illustrated in Fig. 1. The figure highlights the components discussed in the following subsections. Section 5.1 discusses the first component, *suggesting ideas and data discovery*. While Section 5.2 encapsulates *data preparation*, *model development*, and *model evaluation* components in one phase named the development phase. The *model deployment* component is discussed in Section 5.3. The figure shows the standard functionalities in the ML pipeline with their interactions with the business domain. Although in the framework, the business domain supervises the complete pipeline, it interacts with the pipeline in two points. The first point is the *suggesting ideas and data discovery* component. Since the framework is designed to utilize business data in developing B2B predictive systems, data coming from business domain is considered the primary input to the framework. The second point is the *model evaluation* component, as the framework creates a feedback loop from the business domain to the ML pipeline.

The framework was designed according to the Lean Startup methodology [13]. The Lean Startup is an approach that aims to shorten the development process of products and startups. It follows an iterative process of multiple product releases, hypothesis-driven experimentation, and validated learning. Similarly, MLean adopts the same methodology

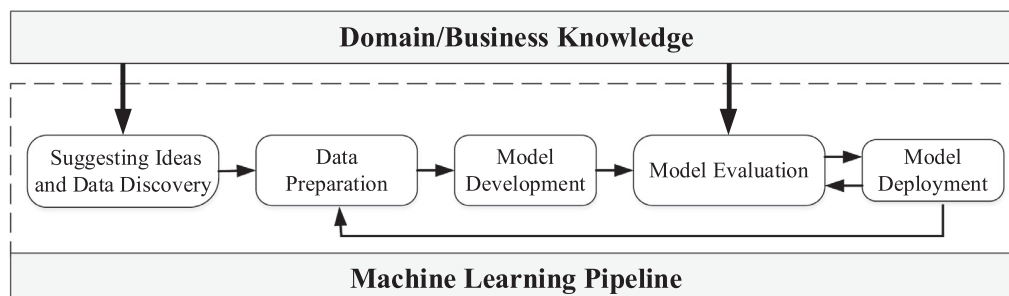


Fig. 1. High-level component overview of the MLean framework.

Table 1
Proposed framework vs. Lean Startup approach.

Main points	Lean Startup	MLean framework
Main Motive	Startups begin with an idea for a product; entrepreneurs think it is fit for the market. However, after development, they fail to reach their customers because they never spoke to a sample of the customers before.	In ML, data scientists build a model, get good results against training data. However, after deployment, the model performance shows severe degradation.
Work Around Uncertainty	Lean Startup methodology eliminates uncertainty by conducting iterative experiments with real customers so the management can continuously check if the market window is still valid. Therefore, Lean Startup methodology can help organizations to test their vision iteratively, and eliminate the uncertainty through the development phase.	Uncertainty in ML originates from different resources: • Model usefulness: MLean eliminates this uncertainty by initiating discussion circles from the very beginning (Section 5.1). • Model performance: MLean eliminates a part of this uncertainty by providing business data as an input to the development phase (Section 5.2).
Eliminate Inefficient Practices	At each development cycle, Lean Startup experiments to validate business hypotheses. So, the management can decide if the product is ready for the market. In the meantime, the experiments help to test the product with real customers and increase consumer awareness.	The framework accommodates user culture through the development phase. Hence, the development team can ensure that the model has prospective users. Moreover, the framework creates a feedback loop from the business domain to the development cycle at each development iteration (Section 5.2).
MVP/ MVM Development	A central module of the Lean Startup methodology is the build-measure-learn loop. In each development cycle, the business develops (build) a minimum viable product (MVP) to begin the experiments (measure). Then, the startup starts working on adding more improvements which depend on the learning process (learn) obtained from the experiments.	The build-measure-learn loop is the core of the development phase. The development team starts the development (build) once the system hypotheses are defined. Each development iteration aims at building a minimum viable model (MVM). Then, the model is evaluated to validate the hypotheses (measure). Based on the evaluation results (learn), the development team can decide on the next steps.
Validated Learning	Validated learning demonstrates the startup progress. Once entrepreneurs adopt the concept of validated learning, they can shorten the development process significantly.	In the framework, validated learning follows the experimental results. If the decision is to Pivot, the hypotheses need to be readjusted to reflect the learning. If the decision is to Persevere, the system is deployed while planning for future improvements.
Continuous Maintenance vs. Continuous Deployment	At some point, the startup stabilizes with successful launches and steady growth. However, the management maintains its success by continuously re-evaluating their vision and running experiments to validate new hypotheses regarding future improvements.	In ML, the model performance degrades once it is put in production. Therefore, the development team must keep monitoring the model performance forever. Once the development team notices a problem in production, the team can step in and fix it.

while considering the challenges in the ML pipeline. Table 1 concludes the main points in the framework and maps them to the principles of the Lean Startup methodology.

5.1. Getting more from business data: ideas suggestions and data discovery

The framework starts with a preliminary phase in which possible opportunities for domain improvements are recognized. In the framework, this is accomplished by initiating discussion circles with individuals in different roles in the organization. Table 2 shows the overall outline of the framework phases. For each phase, the table shows the main objective, the research questions, and the recommended method for data collection. The research questions that this phase aims at answering are listed in the table. Answering the first two research questions (RQ1.1 and RQ1.2) is about matching the right dataset with the right opportunity. Therefore, insights from business executives and data scientists are required to formulate a unified answer for both questions. Also, after defining this match, the impact of the proposed solution on the business

must be evaluated (RQ1.3). Therefore, business leaders should assess the derived business impact to answer the third research question.

Thus, we recommend adopting in-depth semi-structured interviews [29] to collect data in this phase. Semi-structured interviews are guided by topics containing primary questions that must be used in the same way through all interviews. However, this structure allows new ideas to be discussed during the interviews. An important reason to recommend semi-structured interviews is that, in this phase, the framework users already have some understanding of what is happening within the organization. However, they could use open-ended questions in semi-supervised interviews to obtain a deeper understanding and encourage respondents to share their opinions.

The process of data collection is structured in an iterative layered fashion and summarized in Fig. 2. As the figure shows, interviews in Layer 1 aim at obtaining qualitative data about the datasets stored by the organization. The respondent sample in Layer 1 is a group of data scientists and data analysts. We recommend using purposive sampling to choose the respondents sample [30]. More specifically, expert sampling [30] can help in acquiring the knowledge established in the form of

Table 2
Outlines of the framework phases.

Phase	Objective	Research questions	Method for data collection
Phase 1	Exploratory	RQ1.1. What business problem can be solved using ML? RQ1.2. Does the business have enough good quality data to apply ML techniques to solve the defined problem? RQ1.3. Should this predictive model be built? If yes, what are the initial business requirements for this model?	Iterative process of in-depth semi-structured interviews
Phase 2	Improving	RQ2.1. What hypotheses can be derived from the system requirements? RQ2.2. What data preparation activities the development team needs to perform to prepare the dataset? RQ2.3. What are the primary design decisions to build the MVM? RQ2.4. How can user feedback be iteratively incorporated in the model development-evaluation loop?	Indirect methods including job shadowing, observation, and think-aloud protocol
Phase 3	Improving	RQ3.1. What are the thresholds of quality metrics that define the need for retraining the model? RQ3.2. Can automation be adapted to maintain the feedback loop from the real world to the model development team?	Independent analysis using recorded user feedback for models predictions

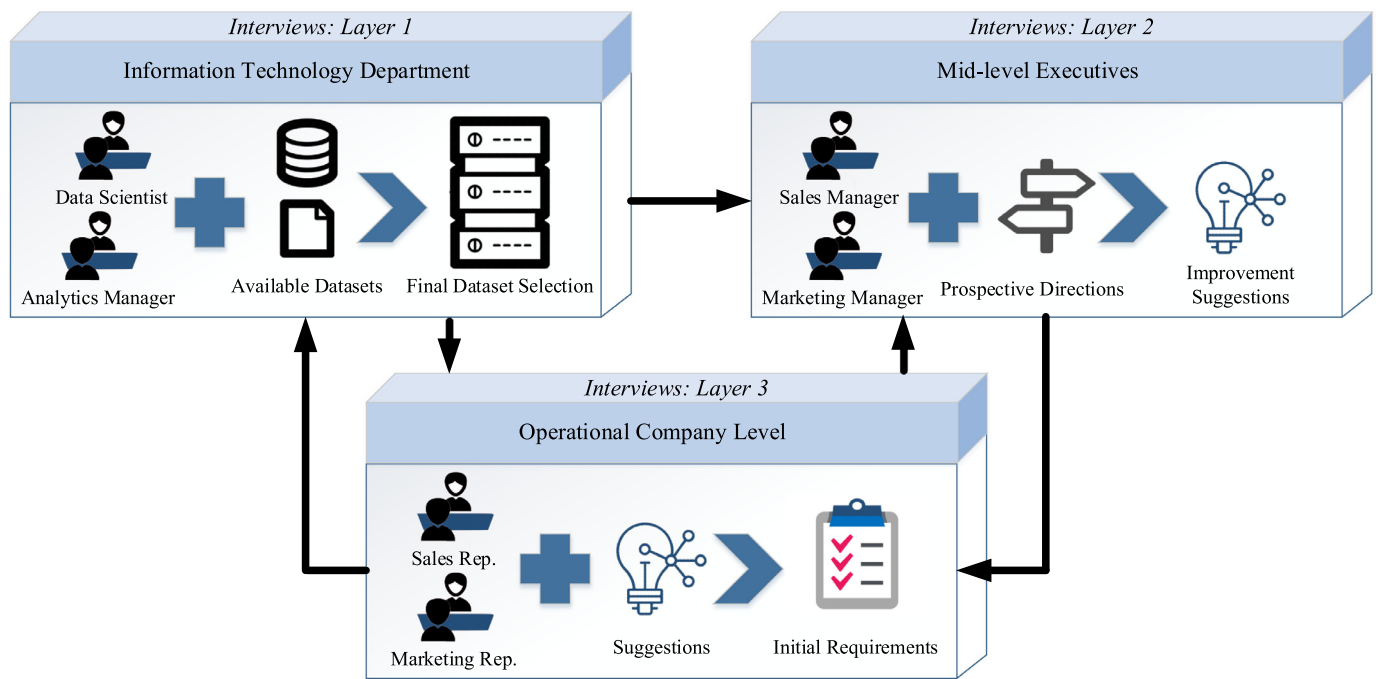


Fig. 2. Interviews structure in Phase 1.

expertise. Based on the findings of the interviews in Layer 1, interviews in Layer 2 can be structured to capture the main challenges the business faces (RQ1.1). The respondent sample in this layer is formed from the middle management of the organization. This layer of interviews could help in refining a list of suggestions for proposed solutions (RQ1.2). These solutions can utilize the datasets (from the interviews in Layer 1) and assist in resolving the challenges in the work process.

After forming a clear understanding of the possible solutions, interviews in Layer 3 can be used to give a qualitative answer to RQ1.3. The respondents in this layer are a sample of the potential end users of the final predictive model. They need to qualify the business value derived from the prospective model and define the essential requirements to maximize its business value [31].

Although requirements elicitation is a crucial task of requirements analysis, we think that using interviews can be beneficial for many reasons. First, there is a high probability that the respondents in this layer will be the end users of the final model. Hence, the interviews can help to estimate the business value derived from the model. Second, many studies [32,33] highlight that, among the existing methods for requirements elicitation, interviews are the most frequently used for determining requirements. Requirements elicited at this point must identify the business preferences regarding the following points:

- Which quality metrics affect the models business value the most? Usually, there is a tradeoff between quality metrics (i.e., a model that achieves a perfect precision value usually has low recall). Therefore, the initial requirements must define user preferences for performance measures used to evaluate the model.
- Which data sources can be used to train the model? On the one hand, input data needs to imitate the same data the domain experts use to come to a decision, but on the other hand, the diversity of the data sources can affect the complexity of the data and the choice of the underlying algorithm. Therefore, the data science team must confirm the availability of the data and its conformity with the rest of the requirements.

As the answers for the three research questions (RQ1.1, RQ1.2, and RQ1.3) must converge to describe one unified system, the proposed process of data collection presented here is iterative. Thus, after eliciting

the requirements for a predictive model, the framework users can go back and talk to the data scientists to validate the compatibility of the dataset with these requirements. The framework users can only exit this phase when they acquire a consistent set of answers for the research questions.

5.2. Developing the solution: data preparation, model development, and evaluation

This section describes the components of data preparation, model development, and model evaluation. A detailed illustration of the phase is presented in Fig. 3.

5.2.1. Data preparation

As shown in Fig. 3, the data preparation phase has two main goals. As for the first goal, it aims at collecting data from the sources identified in the previous phase (Section 5.1) and transforming it into a form that can be used to train a ML model. As mentioned before, data in B2B scenarios are more complex. Hence, data scientists may need to collect data from different unknown sources (e.g., news feeds and social media contents). Therefore, the figure depicts that data sources are not only limited to historical records, but also additional sources can be identified throughout the framework phases. For example, even though some of these sources may be recognized during the first phase, other sources may not be revealed until the evaluation phase, as a part of user feedback.

Moreover, during the process of data transformation, data scientists perform a set of complex activities. Examples of such activities can be summarized as follows:

- **Feature Engineering:** although the term feature engineering may sound related to product line engineering, in ML, the term refers to the process of selecting (and engineering) specific attributes (features) from the input data. It is one of the key activities of data preparation in ML. Data scientists, using the domain knowledge obtained during the last phase, can extract features from the data that are useful for the model to learn.

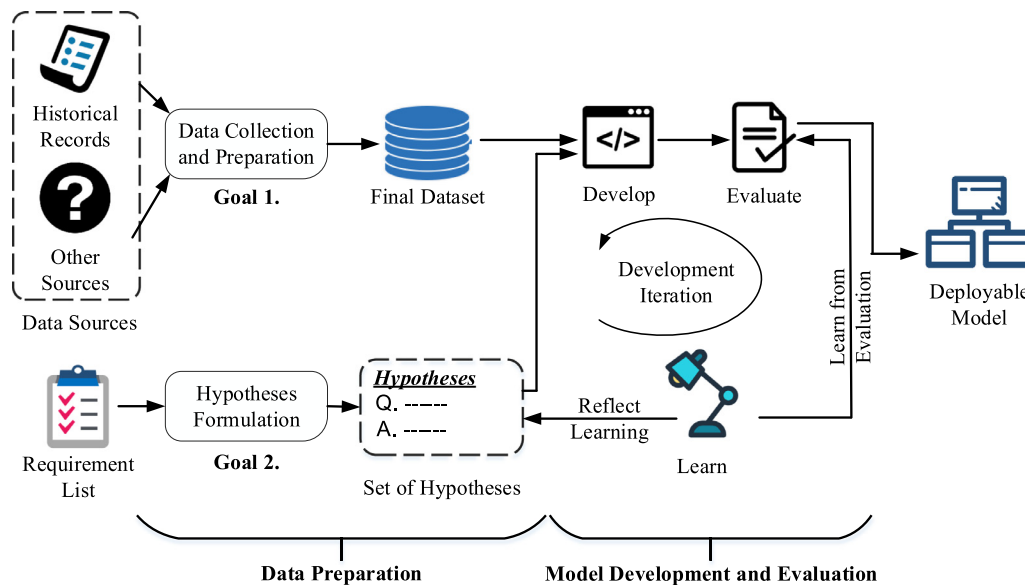


Fig. 3. Data preparation, model development, and evaluation.

- *Ground Truth Generation*: in supervised ML techniques, the model learns from labeled examples. Thus, data scientists need to gather labeled data to train and test the model. Both the size and accuracy of the training data affect the final performance of the model.
- *Deciding on missing values and outliers*: while missing values can compromise the model performance, outliers can affect the model output [34]. Therefore, data scientists need to decide on which method should be adopted when dealing with missing data and outliers.
- *Data anonymization*: even though, our framework aims at creating predictive systems that will be deployed internally within the organization, the process of data anonymization is an essential step before conducting any analytics.

The second goal of the data preparation phase is to formulate a set of hypotheses to define the response of the end users towards the models anticipated behaviors [13]. At this preliminary point, the hypotheses should describe the business value expected from this system, and the performance level that must be obtained before deployment. The hypotheses help the data science team to start building the model as soon as possible. For example, to implement a recommender system, instead of spending time refining the requirements list, the development team can formulate a set of hypotheses from an initial requirements list and start experimenting. The requirements list can be further updated when the users evaluate the model. Adopting the approach of build-measure-learn [13] in this early phase enables the end users to trust the model from the very beginning.

There are at least two main hypotheses that can be formulated at this point. These hypotheses are the business value hypothesis and the performance criteria hypothesis.

- The business value hypothesis tests whether a model can deliver business value to the enterprise or not.
- The performance criteria hypothesis tests if the developed model can meet the performance criteria specified in the requirements list.

As for the performance hypothesis, it is important to note that evaluating ML models differs from evaluating the quality of software products. While evaluating software products considers different characteristics (e.g., ISO 25010) such as efficiency and usefulness [35], evaluating ML model primarily depends on statistical evaluation metrics such as precision and the classification accuracy [36].

5.2.2. Model development

There are two inputs to the development component: the final dataset and the set of hypotheses. The development phase aims at producing an initial model that validates the set of hypotheses. The development phase tries to shorten the development time by applying a set of development iterations. Every iteration involves an experiment to assert the validity of the hypotheses. The first iteration starts by building a minimum viable model (MVM). Then the model is evaluated to validate the hypotheses. The evaluation procedure needs to quantitatively measure the model performance and collect qualitative user feedback as well. A sample of end users, ideally actual end-users but end-user proxies are a viable alternative, must test the prototype as a part of the evaluation process. As in the first phase, purposive sampling [30] can be used to select this sample.

The evaluation results will determine if more development iterations are needed. Terminating the development phase and deploying the model is considered as a business decision that is made after analyzing both the data collected from user feedback and the quantitative data obtained from the models statistical evaluation. Since in the business domain, end users will derive business actions from the models output, users may have specific requirements that must be satisfied in the models final predictions. Therefore, the development team can use the insights collected during each iteration to shape the next development cycle in a way that increases user trust and acceptance.

5.2.3. Model evaluation

After receiving the evaluation results, the development team and business leaders can examine these quantitative and qualitative results to decide if they should pivot (initiate more development iterations) or persevere (terminate the development phase). A pivot is a structure set of corrections to test another fundamental hypothesis. In this case, the set of hypotheses are changed and a new development iteration is initiated. This helps to iteratively validate the business value hypothesis. Alternatively, persevering means that the current set of hypotheses are initially validated, which means that the model is ready for the phase of *continuous deployment*, in which the model is deployed and continuously improved and evaluated to maintain the achieved business value.

The development phase adopts an improvement approach [37]. A list of research questions of the phase along with the overall outline is summarized in Table 2. Since the phase accommodates user feedback to evaluate the model at each (development) iteration, the process of collecting user feedback can be challenging for many reasons. First,

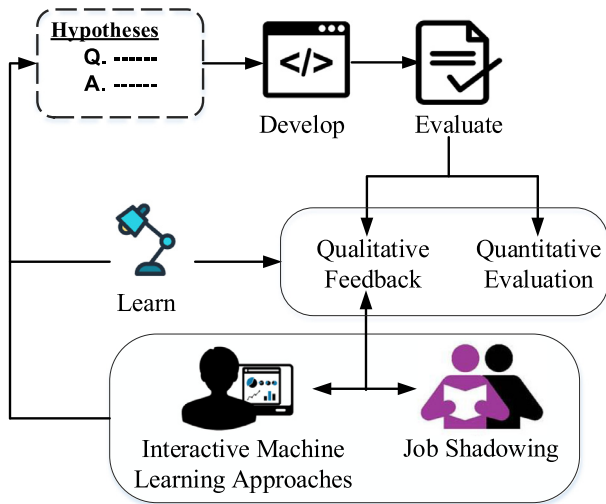


Fig. 4. Cooperating user culture in model evaluation.

collecting qualitative data requires qualitative analysis. Second, analyzing user feedback must consider user culture. For instance, if users think that the automation provided by the predictive system can threaten their jobs, they might not be willing to provide constructive feedback. Thus, the framework users may need to collect data about the internal work process that forms the end user culture.

The process of refining the hypotheses set while considering the user culture is modeled in Fig. 4. The figure shows that the framework can employ two techniques for cultivating user culture. As for the first technique, the framework users can conduct informal interaction, adopting ethnographic research to derive the causes of user behaviors and consider these causes when analyzing user feedback. Moreover, the development team can influence user culture and assure the users that using ML can never replace the need for human creativity. This could be accomplished by demonstrating interactive ML approaches [38] as an

optional design path. The framework users may need to convince the users that they can be involved in evaluating and modifying the model.

5.3. Starting it all over again: model deployment

The deployment phase is designed to preserve model performance achieved in the development phase. One crucial factor that affects model performance is data freshness. Since data in business domain is affected by many factors such as competitors promotions (external) or changes in a business policy (internal), the model performance can drastically vary once put in production [39]. Therefore, the framework requires the model developer to keep evaluating the model with the latest data on a regular basis. The framework treats this re-evaluation step as another set of experiments to validate the system hypotheses. Fig. 5 shows an overview of the deployment phase with its interaction with the development phase. The figure shows that the development team needs to continue collecting business data to re-evaluate the model. The results obtained from the re-evaluation are then used to validate the system hypotheses. If the hypotheses were validated, this means that the users still trust the model. Alternatively, if the evaluation results did not validate the performance hypotheses, this may mean that the performance degradation is caused by a drastic change in the input data. In this case, the development team can investigate this change and reflect this learning on the hypotheses set.

The deployment phase aims at achieving an improving objective with the outlines presented in Table 2. Moreover, by examining Fig. 5, one can spot a repeatable loop of collecting data, retraining, evaluation and deploying. As the loop does not require much designing effort, we recommend that businesses consider automation by creating a validation platform. The framework recommends designing the validation platform to achieve three goals:

- Automatically collection of user feedback;
- Collecting quantitative data about the models performance; and
- Performing statistical estimates of the status of the real world.

By collecting user feedback, the development team can analyze collected data without the need for conducting direct or indirect

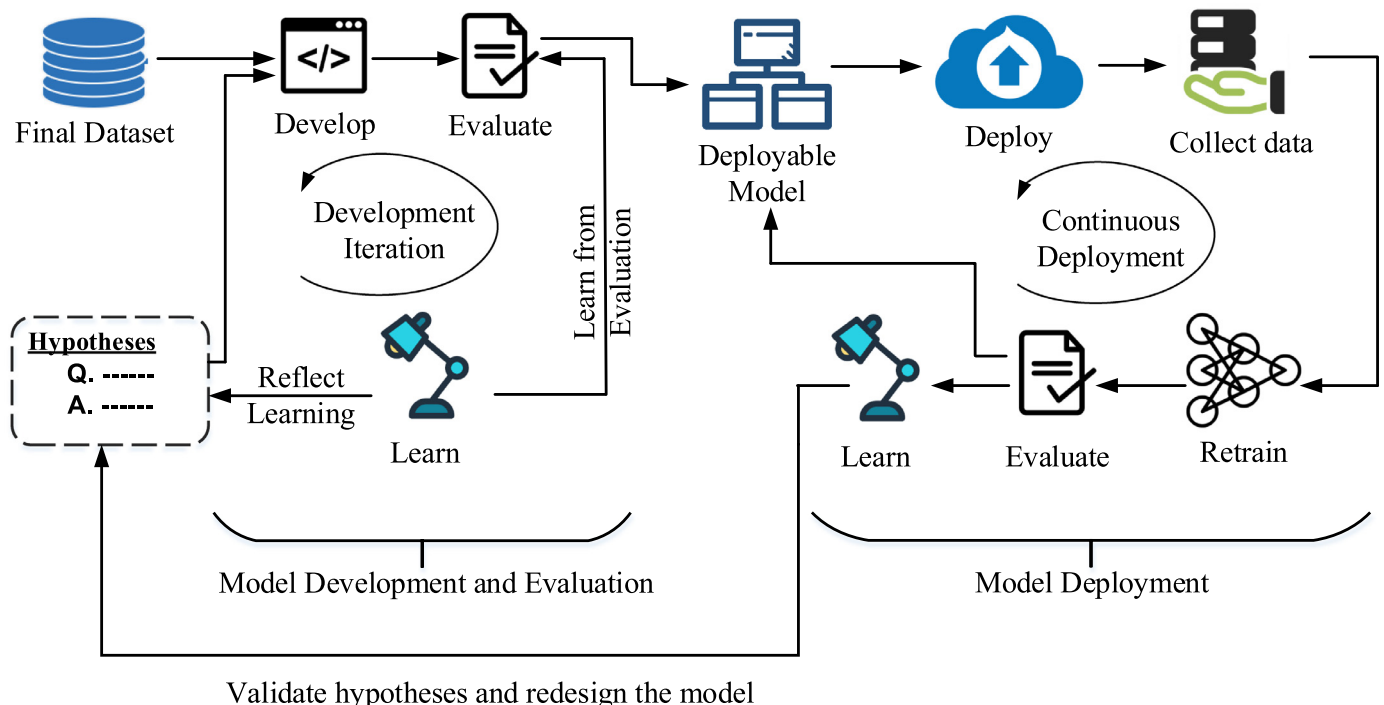


Fig. 5. Model development and model deployment phases.

collection methods. This can decrease the cost of data collection. Also, since the model must be evaluated using new data at frequent intervals, the validation platform can employ visualization assistants to enable business leaders to understand the trends in the evaluation results. Regarding the third goal, it is essential, when accommodating changes in the real world, to consider the changes for which the model is responsible. For example, after deploying a system that recommends clients for marketing campaigns, the collected data will reflect the policies enforced by the system. Therefore, analyzing such data will not consider some blocked changes in the world such as assessing the success of approaching unrecommended clients. Thus, for evaluation, the development team needs an approximation of the distribution of events that would exist in the absence of their intervention (the model). Hence, we recommend designing the validation platform to perform this statistical estimate [40]. One attempt to do such estimates is to deliberately let through some of the blocked events. For example, if the model is recommending clients with a confidence score $P_i(\text{success})$ larger than a threshold (e.g., $P_i(\text{success}) \geq 0.6$), the validation platform can recommend some clients by applying propensity function [40] to choose a set of blocked clients whose confidence score is close to the threshold (e.g., $0.4 < P_i(\text{success}) < 0.6$). Since the model is more uncertain around the threshold value, monitoring the outcome of these recommendations can help to refine the model policy.

To give an example of how to calculate precision and recall in such situations, let us assume that the model examined 1M clients and initially recommended 400,000 clients. Then, the platform recommends a random set of 30,000 blocked clients to the users. After examining the results, the end users reported that only 6000 of this set were true positives (the clients positively responded to the campaign), while the rest (24,000) were false positives (the clients rejected the campaign). At this point, the platform should give each of these allowed events a weight value (W_i), as each event is a representative of the pool of the blocked clients. Therefore, the platform can weigh each allowed sample by $W_i = \frac{1}{P_i}$ which represents a geometric series. Alternatively, for the original 400,000 recommended clients, the results showed that only 4400 were false positives.

Hence, by analyzing the outcome of the allowed events, the model performance can be estimated as follows:

- Negative replies caught by the platform (TN) = $24,000 * \sum_{i=1}^{24,000} \frac{1}{P_i}$
- Precision of the blocking policy = $24,000 * \sum_{i=1}^{24,000} \frac{1}{P_i} / 600,000$
- Recall of the blocking policy = $24,000 * \sum_{i=1}^{24,000} \frac{1}{P_i} / (24,000 * \sum_{i=1}^{24,000} \frac{1}{P_i} + 4400)$

Moreover, the development team can repeat the experiments with different values for thresholds to test many alternative policies.

6. Case study: license cancellation prediction

This section describes the application of the MLean framework to a real-world case study. At each phase, we followed the proposed outlines to answer the research questions.

6.1. Case study settings

In this subsection, we firstly discuss the stakeholder profiles and their interactions during each. We then present the settings of the final model targeted from the framework.

6.1.1. Stakeholder profiles and interactions

IBM is a multinational IT company that provides a range of products and services. We work closely with the Analytics Development team. The team consists of a machine learning architect who leads a team of four data scientists and three data engineers. The teams skills include machine learning, artificial intelligence, and data visualization. The team works to provide data analytics services and build revenue-impacting ML models to different business units in IBM.

During the case study, the research team interacted with three stakeholder groups namely, the analytics development team, a group of mid-level managers, and a group of sales and subscription (S&S) representatives. Fig. 6 shows the three groups and their interactions during each phase. As the figure depicts, the analytics development teams main interactions are in *data discovery* component, *data preparation* component, and *model development* component. In data discovery component, the analytics development team provided information about the organizations stored datasets. While in the development phase, the analytics development team, with their technical experience, directed the data preparation and model development components. As for the *idea suggestion* component, the mid-level management team provided information about the business challenges which ML models can resolve. The management team also played an essential role in the *model evaluation* component, as they supervised the model development phase and decided when to terminate the development phase and deploy the model. The third group of S&S representatives interacted in the *requirements elicitation* component as they helped the research team to define an initial list of goals (requirements). They also evaluated the model during the *model development* and the *model deployment* phases.

6.1.2. The final model settings

With the analytics teams assistance, we applied the framework to an end-to-end situation where we started by identifying possible opportunities for ML, identifying data sources, developing a predictive model for license cancellations, evaluating its outcome, and initiating field tests as a preliminary phase of deployment. Fig. 7 shows an overview of the develop-evaluate-learn cycles followed in the case study. The figure

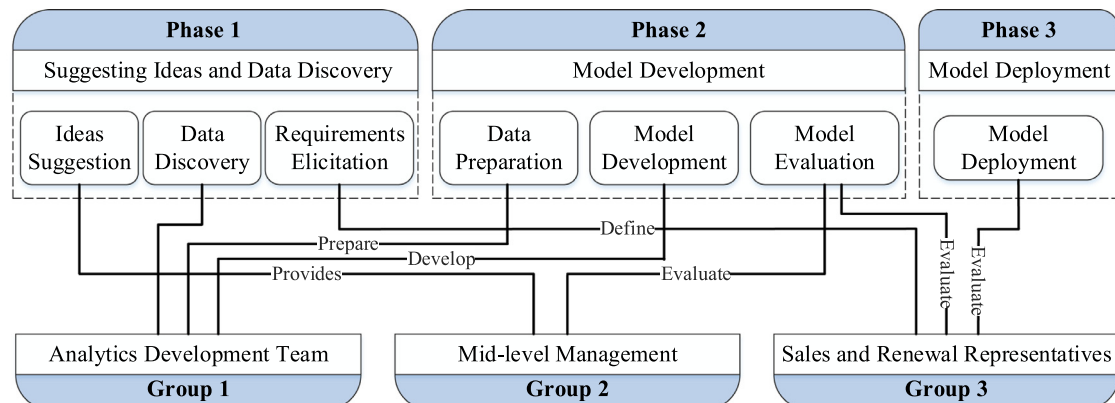


Fig. 6. Stakeholder groups and their interactions.

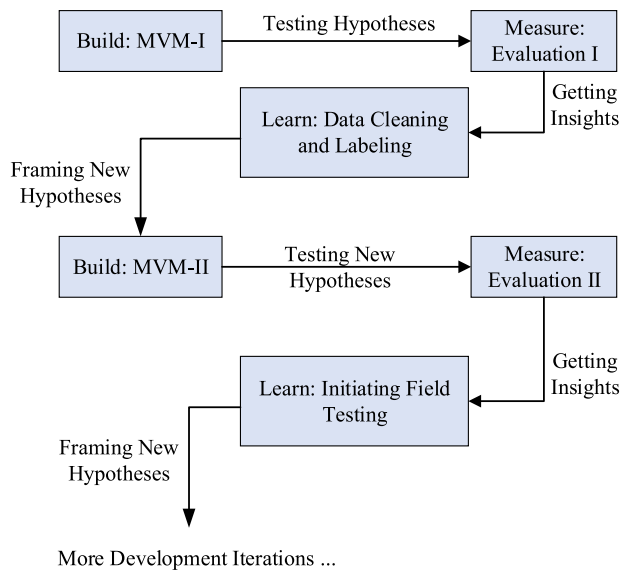


Fig. 7. Develop-Evaluate-Learn cycles.

only shows the executed cycles up to the point this article was written. The final MVM is currently undergoing its first iteration of field testing; hence, unfortunately, no further data could be collected. This MVM was created after running two development iterations and was trained and evaluated using five years of renewal transactions of over 1.3 million purchase orders and 11 attributes. The dataset is commercially sensitive and describes the customer license agreements information. Therefore, we only have limited access to a completely anonymized version of the data. These agreements include conditions on using the features of the purchased software. When the license is about to expire, the customer needs to either renew it by placing a purchase order or cancel it.

6.2. Phase 1: suggesting ideas and data discovery

To start our case study, we began with the phase of ideas suggestions and data discovery (Section 5.1). To collect the data, two researchers followed the iterative process of in-depth semi-structured interviews. Table 3 gives an overview of the application of this process. The table shows the number of interviewees in each layer, the number of interviews conducted with each interviewee, and the duration of each interview in minutes.

All the interviews in this phase were structured according to the funnel model [41]. A sample of the interviews scripts is presented in Appendix A. The interviews were structured to start with open questions and progress to specific ones. During the interviews, all sessions were recorded in an audio format and then transcribed by the two researchers to capture all the details. Anonymized transcripts were then reviewed by a third researcher (the third author) to validate the analysis results. Although transcribing the interviews was time-consuming and can be avoided especially in the first two layers, the research team found it useful, in this setting, for many reasons. Firstly, it facilitated information sharing among the research team. Second, it helped the research team to agree about data interpretation. Thirdly, it formed a source of reference for the research team in the follow-up interviews in later phases.

Table 3

The iterative process of interviews in Phase 1.

Layer	Number of interviewees	Number of interviews	Interview duration (min.)
Layer 1	4	6	50
Layer 2	3	1	30
Layer 3	8	1	30

Lastly, transcripts were useful in interviews in Layer 3, as the research team found coding necessary to define and prioritize user requirements. The process of interviews was conducted as follows:

- In the interviews of layer 1, we conducted weekly group interviews with IBM Analytics Development team for six months. The respondent sample included the team leader, two data scientists, and one data engineer. During these interviews, the research team collected information about the datasets managed by the analytics team. All the datasets that the team explored were anonymized. The list of these datasets and a short description for each dataset are summarized in Table 4.
- In layer 2, the research team had a series of interviews with three individuals from the management. The management sample contained one program director of IT and analytics, one program leader of worldwide S&S business, and one software S&S specialist. During these interviews, information about the renewal process was collected and analyzed at IBM Canada Head Office in Toronto. At this point, one opportunity arose which is to build a predictive model to predict renewal risks. Thus, the answer to RQ1.1 was framed in terms of a top-level function [42] as: *a system is needed to predict and report renewal risks to the sales team beforehand, so the sales team can proactively try to mitigate these risks.*
- In layer 3, the research team utilized the data collected from layer 2 in their domain analysis and started the requirements elicitation process (RQ1.3). The team conducted a set of interviews with eight S&S representatives. During the interviews, the team elicited the initial requirements in the form of goals using the KAOS model [43].
- The initial identification of the functional and non-functional requirements is presented as a generic goal pattern in Fig. 8. The figure shows that, as for the functional goals, the stakeholders stated the following:
 - FR.1. The system must report the renewal risks at least three months before the renewal due date.
 - FR.2. The system must analyze data from different sources, such as purchase records and submitted support tickets, to achieve accurate predictions.
 - FR.3. The model should not only list renewal risks, but it should suggest an action plan of how to mitigate these risks. It is preferred for the model to adopt the prescriptive analytics [44] paradigm and quantify the effect of future decisions to advise on possible results.
 - FR.4. The system should allow the user to give feedback about the predictions.
- Regarding the non-functional requirements, the users stated two soft goals which are shown in Fig. 8 as parallelograms with dashed borders:
 - NFR.1. As for the quality metric that determines the models performance, the stakeholders expressed a business goal of achieving a high level of accuracy. They set a value of 85% as the minimum accuracy value the model must achieve to be considered for deployment.
 - NFR.2. Additionally, they emphasized that a certain level of interpretability is required to understand the reasoning behind the model decisions. Since interpretability [12] of ML models denotes producing predictions that are understandable to the end users, it is considered as an unavoidable requirement in B2B predictive systems.

Furthermore, to ensure that the business has enough data (RQ 3.1.2), the research team initiated a second iteration of interviews in which they had a group meeting with two data scientists and one business analytic engineer from the analytics team. During this interview, both teams decided on using supervised learning algorithms to build the model with the renewal transactions dataset. With the initial requirements and the identification of the input dataset, the research team decided to proceed to the second phase.

Table 4
Available datasets.

Dataset	Description
Renewal Purchase Transactions	The dataset contains anonymous information about the customers entitlements data. Entitlement information includes the purchase date, license type, the expiration date, and information about the purchased product.
Products Download History	The dataset contains information about customers downloads log for each product such as download date and exact time, the number of downloads, and the software license.
Problem Management Reports	The dataset includes support tickets submitted by the customers. It has information describing the tickets lifetime and the conversations between customers and the support team.
Products Allocations and Deallocations History	The dataset contains information related to the products deallocation to different locations (sites). Usually, in B2B scenarios, the customer has multiple business locations. Thus, since the license is given to a specific site, the customer can choose to move some of his licensed products to another location.
Products Evolution	The dataset contains all the information related to product evolution. This is when the product lifecycle comes to an end, and a new version is available.

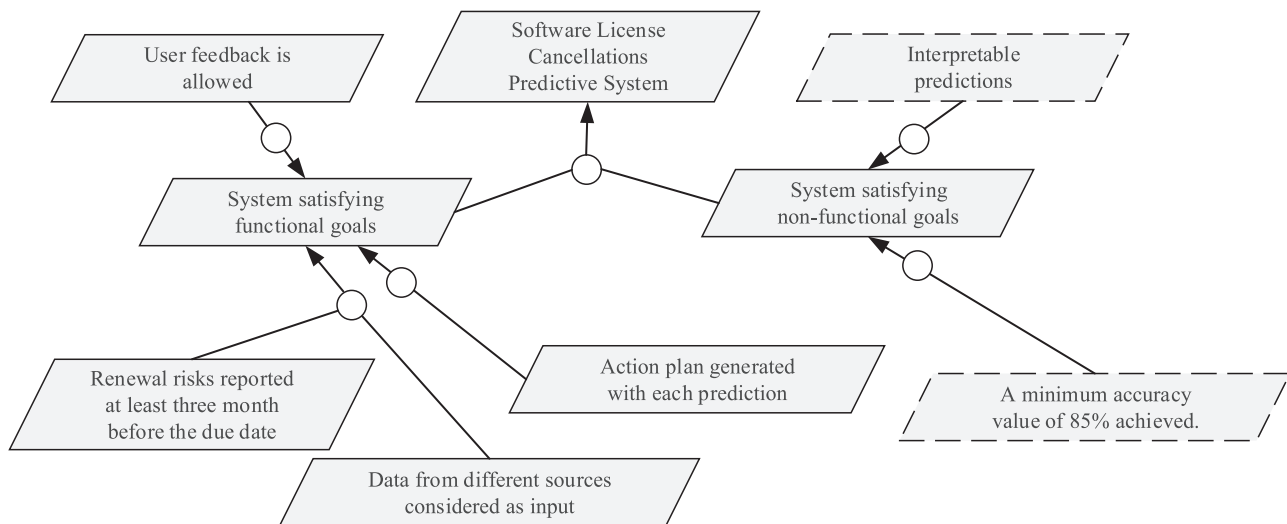


Fig. 8. Generic goal model for the license cancellations predictive system.

6.3. Phase 2 first development iteration

During the data preparation phase, the research team conducted an ethnographic study in which they observe the analytics team while performing the following preparation actions:

- **Feature engineering:** the analytics team employed the domain knowledge gained from the last phase to derive the features. The final set of features has eleven features which can be classified into three categories. The first category includes attributes associated with the clients agreements such as the agreements revenue. The second category contains information about the customers history such as the number of years since the customer purchased his first and last product of the license. The last category comprises a set of aggregated features such as the number of previous renewals and cancellations.
- **Generating ground truth:** since supervised learning algorithms need labeled data [45], the analytics team created a labeling function to automatically label the dataset. The labeling function considered specific attributes in the data such as the end date and the purchase date for two consecutive purchase records. The team used this function to generate ground truth for the data.
- **Deciding on missing values and outliers:** on the one hand, the data analytics team found a small percentage (1.65%) of points with missing values for some attributes and considering the massive volume of the data, these points were ignored by deletion. On the other hand, most of the outliers in the dataset were results of human errors, and hence; were excluded.

- **Data anonymization:** due to the commercial sensitivity of the dataset, the analytics team completely anonymized the data before sharing it with the research team.

Afterward, using the initial requirements list, the research team formulated the following hypotheses list. To speed up the development process, the hypotheses list only focused on a subset of the requirements, more specifically FR.1 and NFR.1:

- **Hypothesis 1:** a model that predicts and reports renewal risks at the beginning of each quarter will add business value (FR.1).
- **Hypothesis 2:** the generated ground truth which is used to build the model is accurate.
- **Hypothesis 3:** a model can be built using supervised learning algorithms and achieve a minimum accuracy value of 85% (NFR.1).

The development of the model was straightforward; the model has a binary output representing the target class as 0 (The customer will cancel) or 1 (The customer will renew). To develop the first MVM, we fed the dataset into multiple supervised ML algorithms: Logistic Regression [46], Random Forest [47], and XGBoost [48]. Each model was trained with a subset of 907,651 records and tested using a held-out test subset of 447,053 records. Table 5 presents the preliminary results of the performance metrics achieved in each case along with the computational time of the model training (in seconds). The performance metrics include Matthews correlation coefficient (MCC) [49], accuracy, precision, and recall. The experiments were conducted on a machine with a Core i7 processor and 32 GB RAM.

Based on the results shown in the table, we selected XGBoost classifier to build the initial MVM, as it achieved the highest accuracy

Table 5

MVM preliminary results in the first development iteration.

Model	MCC	Accuracy	Precision	Recall	Training time (s)
Logistic Regression	0.01	0.73	0.81	0.73	24.40
Random Forest	0.32	0.76	0.79	0.91	41.20
XGBoost	0.33	0.77	0.78	0.96	39.70

Table 6

MVM confusion matrix (Iteration I).

True labels	Total	Non-Renewals	Renewals
Non-Renewals	120,686	31,631 (TN) 26.21%	89,055 (FP) 73.79%
Renewals	326,367	12,994 (FN) 3.98%	313,373 (TP) 96.02%

values (NFR.1). To evaluate the model, both qualitative and quantitative data were collected using statistical validation and user feedback (Section 5.2). The evaluation data was collected as follows:

- **Quantitative data:** the results of the evaluation can be seen in the confusion matrix [50] in Table 6. The table presents True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The table shows that, for predicting non-renewals, the model achieved an accuracy value of 26.21% and a precision of 70.88%. The results attest that the model could not achieve the quality threshold defined in the performance criteria hypothesis (Hypothesis 3).
- **Qualitative feedback:** To gather user feedback, we conducted a follow-up meeting with three S&S representatives and a program leader to review the model output. During the meeting, the research team demonstrated an overview of the dataset and a set of ten predictions produced by the model, along with the quality metrics achieved by the model. Fig. 9 shows a sample of our demonstration regarding the dataset. The figure shows the relationship between the renewals statuses and the number of years since the customers first purchase (Fig. 9(a)) and the last purchase (Fig. 9(b)).

The meeting aimed at involving the users in evaluating both the input and the output of the model. After complete anonymization and aggregation, summarized results were presented to two program lead-

ers and one program director. The meetings findings can be summarized in the following points:

- The end users pointed out that there is a level of inconsistencies between the statistical insights derived from the data and their domain experience. For example, Fig. 9(a) shows an increasing trend in renewals after the first year of purchase, while most representatives stated that most cancellations occur after the first year. Hence, some concerns were raised about the accuracy of the labels, which were generated using a programmatic procedure, in the data.
- The model performance was not satisfactory for end users, as they were not willing to look at a set of reported renewal risks while only 70.88% of them are real problems.

As a result, the research team, the analytics team, and the management team decided to reflect the insights into the hypotheses list and initiate a second iteration of development.

6.4. Phase 2 second development iteration

The insights collected from the evaluation guided the research team to refute the hypothesis of the data quality (hypothesis 2) and pivot to a new direction for generating the ground truth. As a result, a new hypothesis is added as:

- **Hypothesis 2:** Generating the ground truth must exploit domain expertise to ensure the validity of the labels and enhance the model accuracy.

To test the new hypothesis, the research team reviewed the data preparation component, as a new set of labels are required. The team concluded that for any labeling technique that is applied to the business domain, it is required to find a midpoint between labeling accuracy and labeling cost. To find this mid-point, the research team applied the hybrid method from their previous work [51] to generate the ground truth. The final dataset along with the updated ground truth is then used to train the XGBoost classifier. The generated model was evaluated as follows:

- **Quantitative data:** The results after using the updated ground truth are presented in Table 7. The table attests that the model

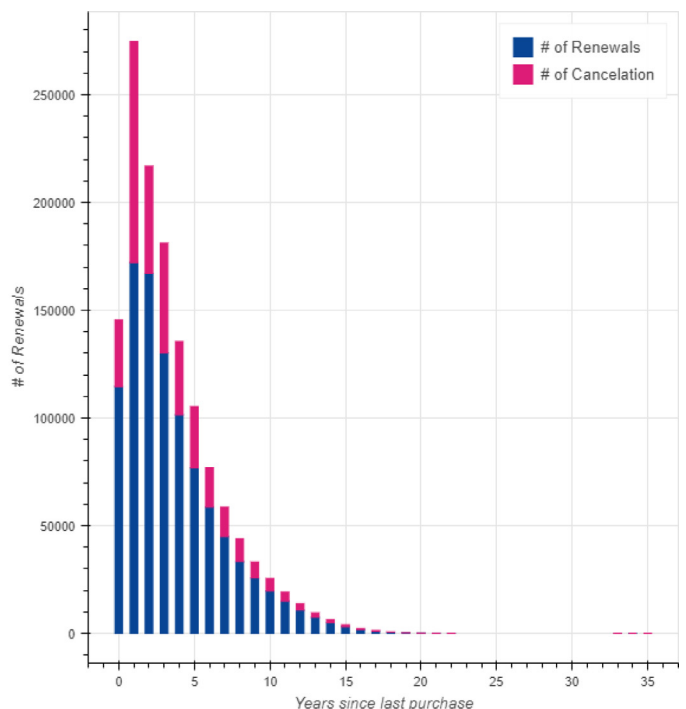
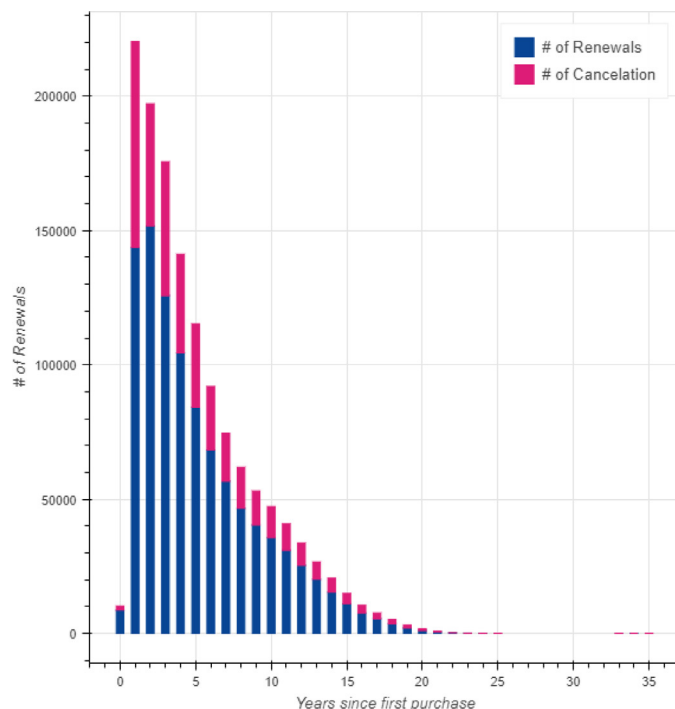
**Fig. 9.** Demonstration of the models input.

Table 7
MVM confusion matrix (Iteration II).

True Labels	Total	Non-Renewals	Renewals
Non-Renewals	120,686	68,197 (TN) 56.51%	52,489 (FP) 43.49%
Renewals	326,367	3450 (FN) 1.06%	322,917 (TP) 98.94%

accuracy increased by 116% and 3.05% in predicting non-renewals and renewals respectively compared to the last development iteration. Moreover, the model performance validated the third hypothesis by achieving an accuracy value of 0.87 and MCC value of 0.67.

- **Qualitative feedback:** The research team had a group meeting with four S&S representatives and two program leaders to review the model output. Since the end users were involved in creating the ground truth [51], they trusted the input data. The meeting focused on reviewing a set of ten predictions presented by the model. Overall, the model performed well in predicting the license cancellations. However, the end users suggested some enhancements regarding the models interpretability (NFR.2).

With these findings, the management team and the analytics team decided to persevere. The persevering decision had two main aspects. Firstly, the teams decided to start a phase of field testing to deploy the model and revalidate the hypotheses with live data. Secondly, the teams decided to run a third iteration of development in which the hypotheses set are updated to accommodate future improvements.

6.5. Phase 2 plans for the third development iteration

Field testing aims at evaluating the model with live data taken from the business domain. Although at this point, we have a model that has validated a partial set of its requirements (Fig. 8), the framework aims at pushing the MVMs into production as soon as possible. Running the model with live data has its advantages. First, it gives a real perception about the model performance. Second, it allows the development team to monitor performance degradation at the same time they are working on future improvements. This can guide them to formulate new hypotheses that need to be validated. Finally, starting the field testing in parallel with conducting further development iterations can speed up model development.

To initiate the field testing, the teams took the following steps:

- The model was deployed internally in IBM with a re-evaluation period of three months.
- At the beginning of each quarter, the model will be used to predict renewal risks. During the quarter, S&S representatives will review the predictions and give feedback to evaluate each prediction.
- At this point, a sample of the end users is chosen to engage in field testing. The sample includes two program leaders of worldwide S&S business and four S&S representatives.
- At the end of the quarter, the quarter data of renewal purchase transactions along with the user feedback will be used to create a status report about the model performance for re-evaluation.
- Unfortunately, automation could not be fully adopted at this point. However, the development team decided to schedule a job on the production server to collect archival data that includes:
 - the data of renewal purchase transactions during the last quarter; and
 - the user feedback recorded during the quarter.
- This data will then be used to evaluate the model performance and decide if the model needs retraining. As mentioned in Section 5.3, this process of continuous monitoring should run forever to prevent performance degradation.

Alternatively, the development team has formulated the following hypotheses for the next development iterations:

- **Hypothesis 4:** considering the client submitted support tickets along with the purchase records in the models input will enhance the model accuracy (FR.2).
- **Hypothesis 5:** providing the models rationale for the generated predictions will enhance the end user trust (NFR.2).
- **Hypothesis 6:** providing an action plan along with each prediction will enhance the models business value and guide the users to optimized solutions to save the renewals (FR.3).

As we were writing this article, the development team was about to start their first round of field testing. Hence, no further data could be collected at this point. As for our next steps, the development team plans to proceed with the third iteration of development to validate hypothesis 4. The team is currently preparing the problem management reports dataset (Table 4) to add it to the input data. All in all, the parallel process of conducting field testing along with initiating more development iterations will be repeated as needed to optimize the models compliance with the system requirements.

7. Discussion and threats to validity

This section firstly discusses the results and lessons learned from the case study. Secondly, it lists the threats to validity.

7.1. Discussion

Our research seeks to build an end-to-end framework for developing B2B predictive systems. During the application of the MLean framework, the results validated our hypotheses (Section 4). As for the first hypothesis (H1), interaction with business leaders and end users helped to shorten the development time. Without them pointing out the inaccuracies in the input labels during the first development iteration (Section 6.3), it would take the development team longer time before considering reviewing the ground truth. Also, regarding the second hypothesis (H2), the results acquired during requirements elicitation show that business requirements are essential for defining the business value of the model and the statistical metrics used to validate its performance. Defining these business constraints, in the beginning, helped the team to shape the development iterations to increase user trust and acceptance. In the following subsection, we discuss the lessons learned from the case study through the main principles of action research methodology [52].

7.1.1. Examining problem features

The MLean framework is designed to be applicable in other organizations that match firmographics variables specified in Section 3, given that these organizations have enough data to apply ML. Also, the framework assumes that the organizations are serious about their analytics transformation which implies that the data science teams in these organizations are actively engaged in collecting masses of data from various sources and developing models to serve different business units. Another hypothesis the framework assumes about the organization is about the technical background of its data science team, as the framework requires the analytics team to have experience in requirements engineering, data analytics, and ML techniques.

Moreover, before applying ML techniques, organizations should do a cost-benefit analysis to estimate if the potential business value is worth the implementation cost. In our case study, this was accomplished during the first phase (Section 6.2), when mid-level management stated that, for large multinational IT companies like IBM, renewal of software licenses contributes to the selling organizations revenue. Therefore, utilizing ML to anticipate the renewal risks is expected to have positive business value to the sales unit.

7.1.2. Practitioners commitment

The application of the framework requires a long-term commitment from the organization and the participants. As for the organization, we

Table 8
Overhead cost for applying the MLean framework.

Stakeholder Group	Number of individuals	Phase	Hours per individual	Total
Sales and Subscription Representatives	8	Phase 1 (Requirement Elicitation)	1.0	8
	3	Phase 2 (Development Iteration I)	2.0	6
	4	Phase 2 (Development Iteration II)	1.5	6
Mid-level Managers	3	Phase 1 (Idea Suggestion)	1.0	3
	1	Phase 2 (Development Iteration I)	2.0	2
	2	Phase 2 (Development Iteration II)	1.5	3
Analytics Development Team	4	Phase 1 (Data discovery)	6.0	24
	2	Phase 2 (Development Iteration I)	2.0	4
Research Team	2	Phase 1 (Transcribing and Analyzing Interviews)	46.0	92

found applying the framework, in many ways, similar to committing to building a product, except in this case the product is a B2B predictive system. According to the report from McKinsey institute [53], organizations commitment to ML is empowered by the benefits anticipated from the output systems. Moreover, since building user trust and accommodating user feedback is essential to create a successful B2B predictive model [12], in our case study, the organizations commitment to the MLean processes was trivial to sustain a successful product line.

Also, as for participants commitments, it is true that applying MLean adds an overhead cost to the costs associated with the standard application of ML (e.g., cost of data preparation). This overhead cost is related to data collection and analysis processes during the framework phases (e.g., conducting and transcribing interviews). To quantify these costs in the form of personal effort, Table 8 shows a detailed cost breakdown of our case study. The table shows that the groups that bore the highest costs were the research team and the analytics team. As for the research team, the costs came from transcribing the interviews. Although we found transcribing necessary in this setting, this cost could be significantly reduced if transcribing is skipped in the first two layers (as mentioned before in Section 6.2). Alternatively, as for the analytics team, a total of 24 h of interviews were logged during the data discovery phase (Phase 1). However, most of this time was to inform the research team about the datasets stored in the organization. Therefore, it is expected for this time to decrease when an organization applies the framework without consultancy. Also, based on the feedback collected throughout the case study, we think that these costs are justified by the outcome which is a MVM that was iteratively developed and evaluated to reflect domain knowledge about the renewal process.

7.1.3. Cyclical process model

It can also be seen that, in our case study, the research team played the mediator role during the first phase (Section 6.1). Since the phase involves iterating through different layers of interviews, there must be a team whose responsibility is to coordinate between the stakeholder groups and ensure that all the stakeholders are actively participating. As mentioned in Section 4, the research team was integrated with the data science team; and hence, their roles can be interchanged by the analytics team. Moreover, we think that the mediator role can also be traded by applying some performance management strategies such as balanced scorecard [54]. Balanced scorecards can be designed according to the business needs to accommodate the business goals targeted from each phase, then the management team can determine the best way to achieve these needs. In this manner, the mid-level managers can keep track of the execution of framework activities throughout its phases and monitor the outcomes arising from each layer.

7.2. Threats to validity

Regarding the threats of validity [55], we list the first external validity threat as the possible lack of generalizability of the framework to other organizations since the framework was designed and evaluated using a partnership with only one organization. To mitigate this

threat, we specify the firmographic variables of the proposed framework (Section 3). Moreover, we isolated the design specifications in each phase from the phase outline to allow the users to explore other methods for data collections. Another threat to construct validity [55], is related to the mapping of the data collected during each phase of the framework to the thematic codes. To mitigate that threat, we applied different types of data triangulation [28] to increase the precision of the results. First, we combined different methods of data collections. Second, we collected data from different sources (i.e., interviewees with different roles in the organization) on different occasions (i.e., collecting data at fixed points). Moreover, we shared our thematic codes with the IBM Analytics team and the management sample through general discussions to validate the data mappings. Finally, the application of the proposed framework with IBM lasted over nine months which helped in creating a continued interaction with the participants and validating the collected information at different temporal points.

8. Conclusions

In this paper, we present an end-to-end framework for developing B2B predictive models. We designed the framework following the lean startup methodology to accommodate the business domain and the end-user perspectives. The proposed framework consists of a set of phases during which the framework users can define possible opportunities for machine learning, develop, evaluate, and deploy machine learning systems. The framework presents the predictive models as a new class of problem in requirements engineering. Thus, it proposes a generic approach to organizations to define and elicit requirements for such models. To evaluate the framework, we have undertaken a case study to which we applied the framework to develop a predictive model within our industrial partner IBM. The framework was used to identify the initial requirements of the final model. To develop the model, we performed two development iterations. At the end of each iteration, the model was evaluated using both statistical validation and user feedback. The first iteration introduced having appropriate training labels as one of the most critical systems requirements for machine learning models in the business domain. Additionally, the case study originated a minimum viable model to detect risks of license cancellation. The model is currently undergoing field testing. The results of the case study attest that the application of the proposed framework can help organizations to utilize their stored datasets to effectively build predictive models.

Conflict of interest

None.

Appendix A. Interview Guidelines and Scripts

The MLean framework uses an iterative process of in-depth semi-structured interviews as a method for data collection in the first phase. As mentioned in Section 6.2, semi-structured interviews are organized as

Table A.1
Common questionnaire for interviews in Phase 1.

Layer	Topic	Questions
Layer 1	Overall view of the stored datasets	• What are the datasets managed by your team? • Who are the users for these datasets? • Can you elaborate on the database schemas for these datasets?
	Using ML with datasets	• Have any of these datasets been used in ML models within the organization? • What ML algorithms that are mostly applied by your team?
	Quality of the datasets	• Are the datasets well structured? • How reliable are the sources for these datasets? • How consistent are the datasets?
Layer 2	Overall business processes and business challenges	• What are the challenges that face your team? • In your business unit, do you have any business decisions that can be automated? • What data sources do people in your team usually search for? • Do you apply any automation technique to your customer interactions?
	Reasons and expectations for ML	• What are your expectations for using ML in business processes? • What are the metrics if predicted would have a positive impact on your team? • What are the useful and challenging aspects of predictive systems? • Do you think ML is the right approach to help the business process to be more efficient?
Layer 3	Overall Process of License Renewals	• Can you elaborate on the overall process of the license renewals? • What daily activities related to the renewal process that you do as a part of your job?
	Anticipating renewal risks	• What software systems do you use in your job? • Do you look up the customers renewal history before contacting him/ her? • What kind of information do you use to anticipate renewal risks? • Do you use recommendations from predictive modes to anticipate renewal risks?
	Eliciting Requirements	• What are the most important business requirements for a license cancellation predictive system? • What are the key business data requirements for this system? • How will these requirements help to add value to your team?

a set of open-ended questions and depending on the answers of the interviewee to a given question, the follow-up question can change between different interviews. Nonetheless, once the interviewee set is identified, a set of common questions can be created for all the interviews. The topics of the questions in each layer were derived from the three research questions of this phase, discussed in Section 5.1. The questionnaires that were used to guide the interviews in each layer is presented in Table A.1.

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